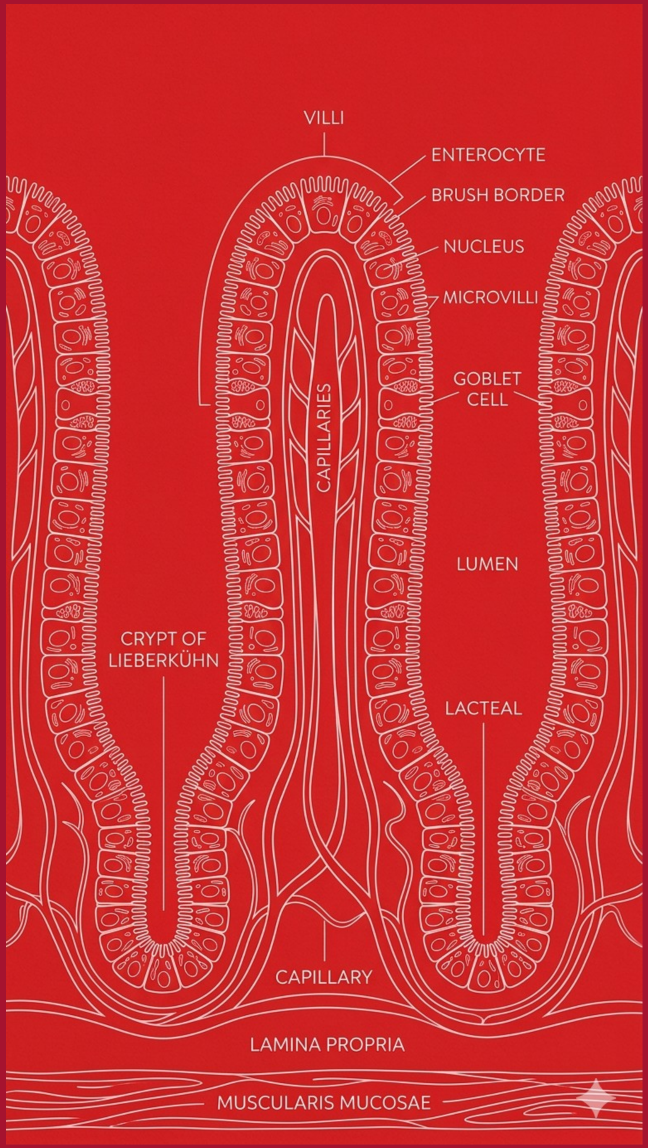
BIOL 40C

Human Anatomy & Physiology III

Class 4 — Absorption, Large Intestine, and Integration



Spring 2026 | Foothill College

Giorgio Lagna, Ph.D. | *Wednesday, April 15, 2026*

Today's Journey

1. **Accessory Organs** — 10:00 – 10:25
   Pancreas, liver, gallbladder
2. **Absorption in Action** — 10:25 – 10:55
   Every transport mechanism at once
3. **Large Intestine** — 11:05 – 11:20
   Water, microbiome, compaction
4. **Integration Capstone** — 11:20 – 11:50
   Coffee with Cream and Sugar

By the End of Class, You Should Be Able To…

CONCEPT

1. Explain how the pancreas, liver, and gallbladder contribute to chemical digestion.
2. Trace the CCK signaling pathway and name three targets it coordinates.
3. Identify the transport mechanism used for each major nutrient at the apical and basolateral membranes of the enterocyte.
4. Distinguish the hepatic portal and lacteal/lymph absorption routes and predict which route a nutrient takes.
5. Describe the functions of the large intestine, including the microbiome.
6. Trace a multi-component meal from ingestion through absorption, naming structures, enzymes, and transport mechanisms.

BLOCK 1

Accessory Organs

Pancreas, liver, and gallbladder — the chemical factories

Warm-Up: Connecting to Monday

ACTIVITY

Quick Write (1 minute)

Name the three levels of surface area amplification in the small intestine and explain why surface area matters for digestion.

→ Then share with a neighbor (30 sec). We will debrief.

The Pancreas: An Enzyme Factory

CONCEPT

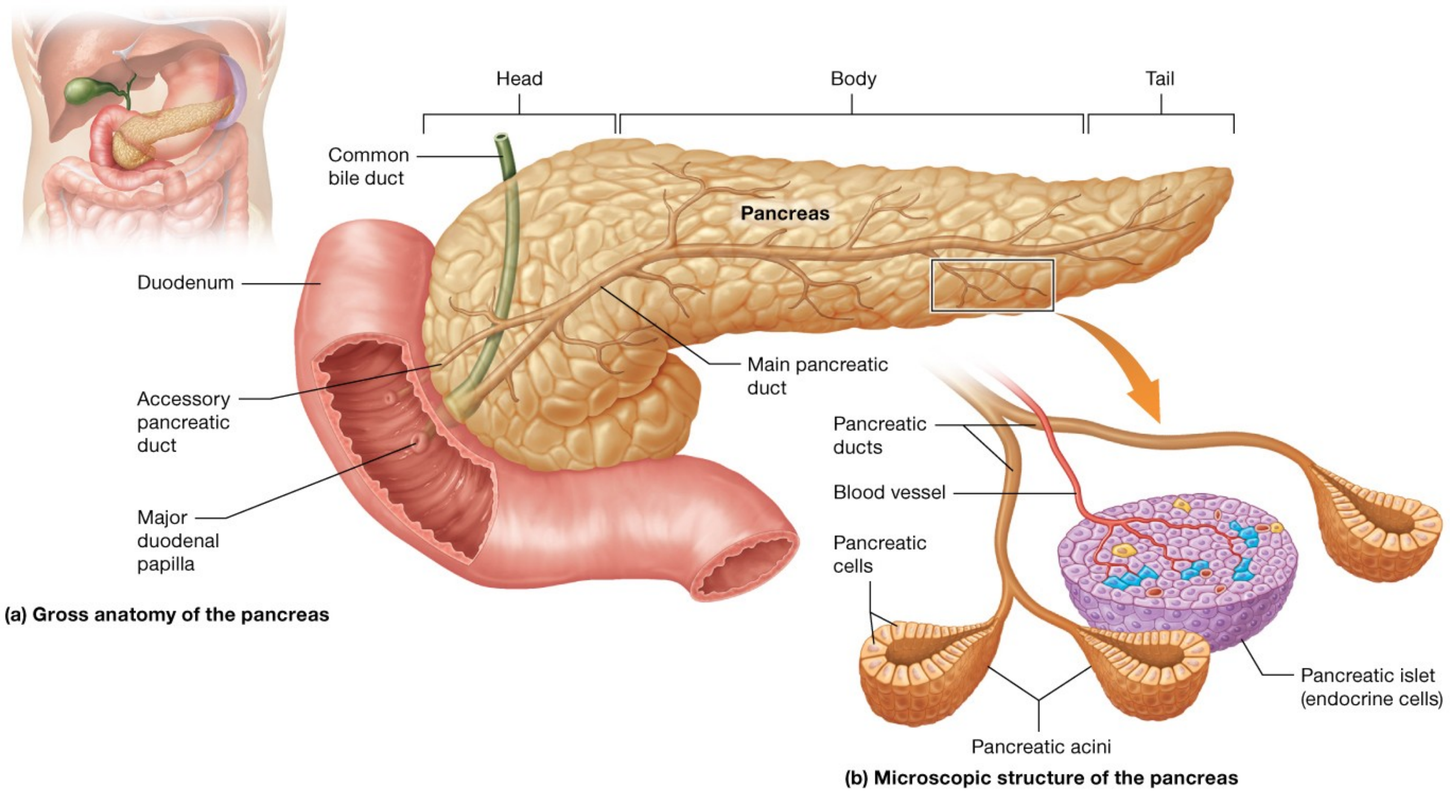
Anatomy

- Head: nestled in the C-curve of the duodenum
- Body: central portion, crosses the spine
- Tail: extends toward the spleen

Dual-role gland:

- **Exocrine** → pancreatic juice into the duodenum
- **Endocrine** → insulin / glucagon into the blood

Today: exocrine side.



Pancreatic Juice: Two Jobs in One Secretion

CONCEPT

1. Neutralize stomach acid

- Bicarbonate (HCO_3^-) raises duodenal pH from ~2 to ~7
- Protects enterocytes from acid damage
- Creates the optimal pH range for pancreatic enzymes
- *Clinical hook: when bicarbonate secretion fails → duodenal ulcers.*

2. Digest everything

- Pancreatic amylase → starches
- Trypsin / chymotrypsin / carboxypeptidase → proteins
- Pancreatic lipase → triglycerides
- Nucleases → DNA / RNA
- Proteases are secreted as **zymogens** *(inactive)*. *Why?*

Concept Check: Why Zymogens?

CHECK

Why does the pancreas secrete trypsin as the inactive zymogen trypsinogen, rather than as active trypsin?

A) Zymogens are cheaper to produce than active enzymes.

B) Trypsin is too large to fit through pancreatic duct cells unless folded inactively.

C) Active trypsin would digest the pancreas itself before it ever reached the duodenum.

D) The duodenum will not absorb trypsin unless it arrives in the inactive form.

Why does the pancreas secrete trypsin as the inactive zymogen trypsinogen, rather than as active trypsin?

0

Zymogens are cheaper to produce than active enzymes.

0%

Trypsin is too large to fit through pancreatic duct cells unless folded inactively.

0%

Active trypsin would digest the pancreas itself before it ever reached the duodenum.

0%

The duodenum will not absorb trypsin unless it arrives in the inactive form.

0%

Answer: Self-Protection

CHECK

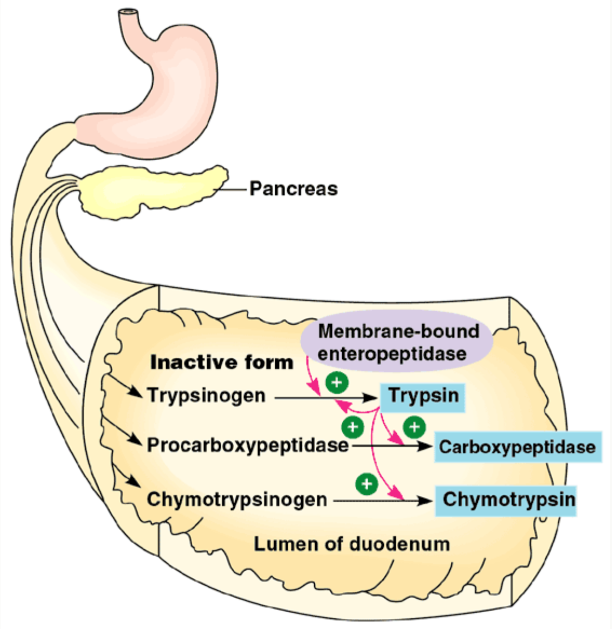
✓ C — Active trypsin would digest the pancreas itself.

The logic: pancreatic acinar cells make proteases. If those proteases were active inside the cell, they would cleave cellular proteins on the spot: autodigestion!

The fix is a two-step activation:

1. Pancreas secretes trypsinogen (inactive) into the duodenum.
2. Brush-border enterokinase (a duodenal enzyme) cleaves trypsinogen → trypsin.
3. Active trypsin then activates chymotrypsinogen and procarboxypeptidase.

Clinical tie-in: acute pancreatitis = premature zymogen activation → the pancreas starts to digest itself. This is one of the most painful conditions in medicine.



The Liver: Over 500 Jobs, But Today Only One

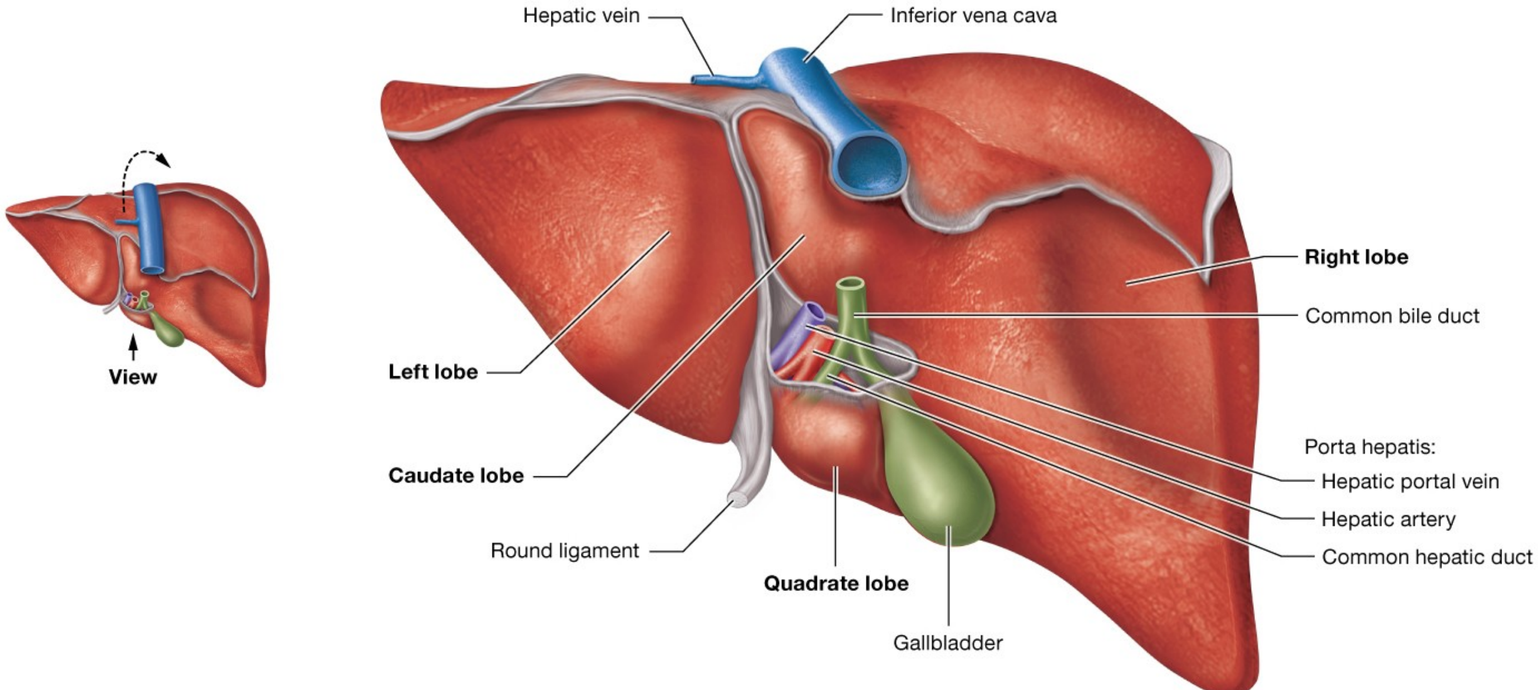
CONCEPT

For digestion: the liver makes bile.

- Largest internal organ (~1.5 kg)
- Four lobes; dual blood supply (portal vein + hepatic artery)
- Hepatocytes continuously secrete bile into bile canaliculi → hepatic ducts → common hepatic duct

Other liver jobs (later in the quarter):

processing absorbed nutrients, detoxification, drug metabolism, plasma protein synthesis, clotting factors, vitamin storage...



(b) Posteroinferior view

Bile: What It Is and What It Does

CONCEPT

Composition

- Bile salts (the working part)
- Bilirubin: recycled from old hemoglobin
- Cholesterol
- Phospholipids
- Water and electrolytes

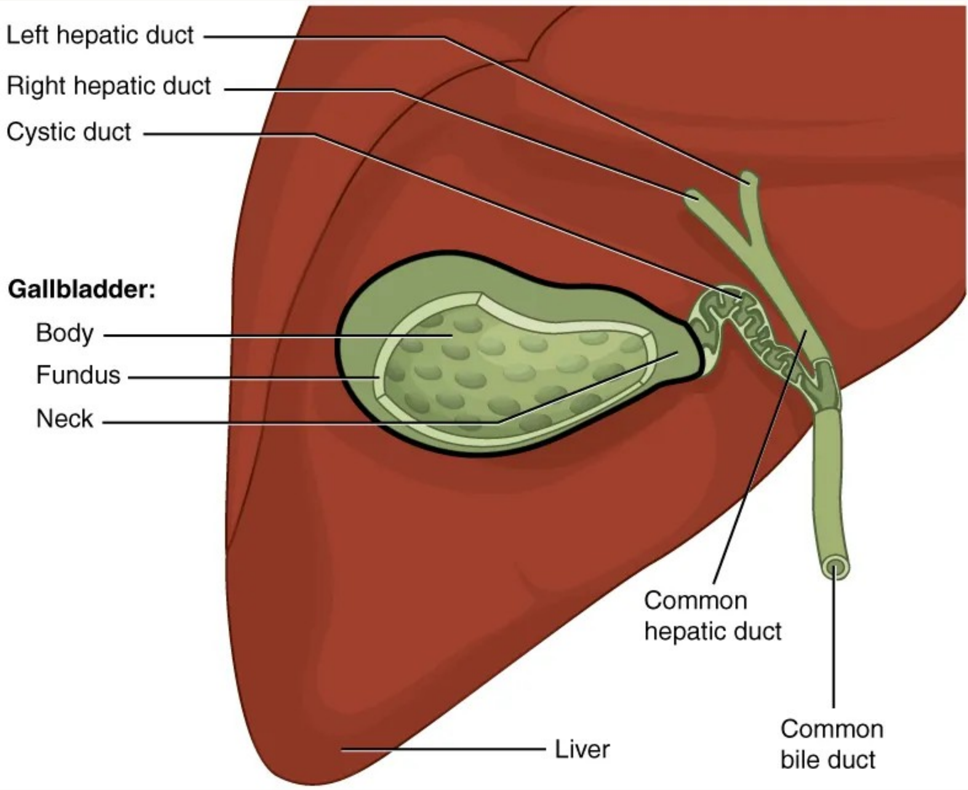
Function: emulsify fats

- Bile salts have a hydrophilic + hydrophobic face
- They coat large fat globules and break them into tiny droplets
- Tiny droplets = more surface area for pancreatic lipase
- **Key idea: bile is NOT an enzyme. It is a detergent that sets the stage for lipase.**

The Gallbladder: Storage and On-Demand Release

CONCEPT

- Stores bile between meals
- Concentrates bile ~10× by absorbing water
- Releases bile when fatty chyme enters the duodenum
- **Release is triggered by CCK (cholecystokinin) from duodenal enteroendocrine cells.**
- *Clinical: gallstones can block the cystic duct → pain after fatty meals → ultimately cholecystectomy.*
- *After removal: bile drips continuously → fats still digestible, but large fatty meals may cause trouble.*



Ligand-Receptor Signaling — The Big Picture

A&P REVIEW

From BIOL 40A: cells talk to each other with chemical messengers.

A ligand (messenger) binds a receptor (protein). The receptor changes shape. The shape change triggers an intracellular response.

1. **Signal** — *A ligand is released*
2. **Travel** — *Through blood or interstitial fluid*
3. **Bind** — *Fits a receptor on a target cell*
4. **Respond** — *Target cell changes behavior*

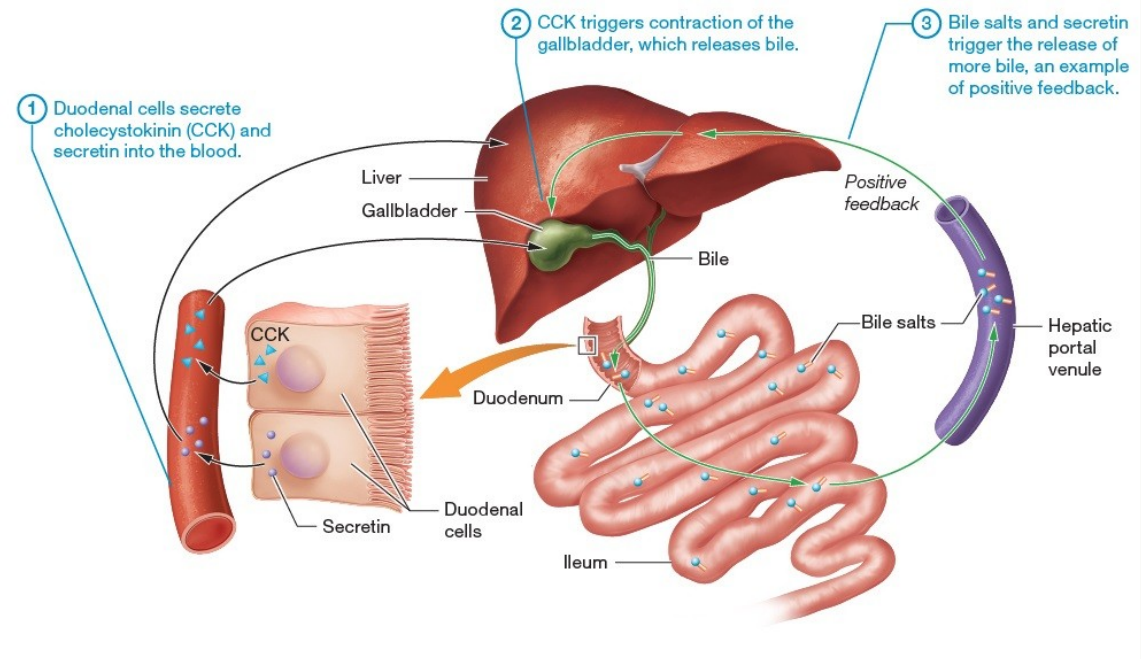
CCK: Coordinating the Duodenum's Response to Fat

CONCEPT

1 **Fat arrives** Lipids enter the duodenum from the stomach

2 **Detect** Enteroendocrine cells (I cells) sense fat/protein

3 **Release CCK** CCK secreted into the blood



4 **Act on 3 targets** Gallbladder contracts • Pancreas secretes enzymes • Hepatopancreatic sphincter relaxes

One ligand → three coordinated organ responses. This is hormonal control in action.

This Pattern Repeats All Quarter

CONCEPT

Today you have seen ligand-receptor signaling coordinate organs across distance.

When we meet the next systems, you will recognize the same logic:

CCK	Gallbladder + pancreas + sphincter	*Week 1 (today)*
ADH	Kidney collecting duct	*Week 3 — Urinary*
Aldosterone	Kidney tubule Na^+ channels	*Week 4 — Fluid balance*
Insulin / Glucagon	Liver, muscle, adipose	*Week 7 — Endocrine*
Estrogen / Progesterone	Uterus, hypothalamus	*Week 8 — Reproductive*

BLOCK 2

Absorption in Action

Every transport mechanism at once

All Transport Mechanisms, At the Same Table

A&P REVIEW

Simple diffusion

Lipophilic molecules cross the phospholipid bilayer (fatty acids, O_2, CO_2)

Facilitated diffusion

Channel or carrier protein; down the gradient; no ATP (GLUT transporters)

Primary active transport

Pump uses ATP directly (Na^+/K^+ ATPase)

Secondary active transport

Co-transporter rides a gradient set up by a primary pump (SGLT1)

Osmosis

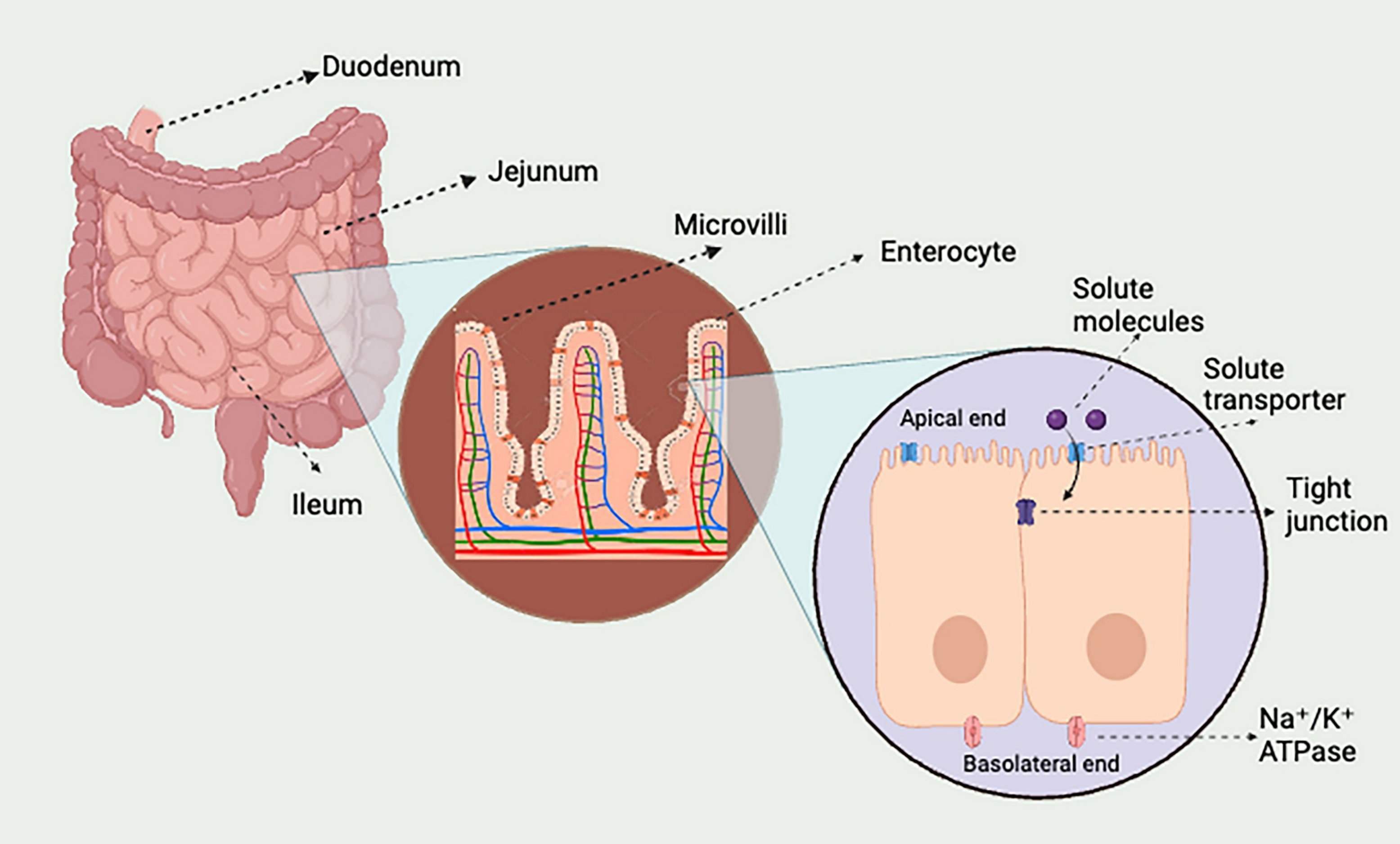
Water follows solutes through aquaporins

Endo / exocytosis

Vesicle-based; for very large molecules (vitamin B12, chylomicrons)

The Enterocyte: One Cell, Two Faces

CONCEPT



Two membranes, two rule sets:

Apical membrane

- Faces the lumen
- Brush border (microvilli)
- **Entry transporters**

Basolateral membrane

- Faces the blood / lymph
- **Na^+/K^+ ATPase** lives here
- **Exit transporters**

The Na^+ Gradient Is the Engine

CONCEPT

Example: carbohydrate digestion and absorption in the small intestine.

One pump powers the whole absorption system.

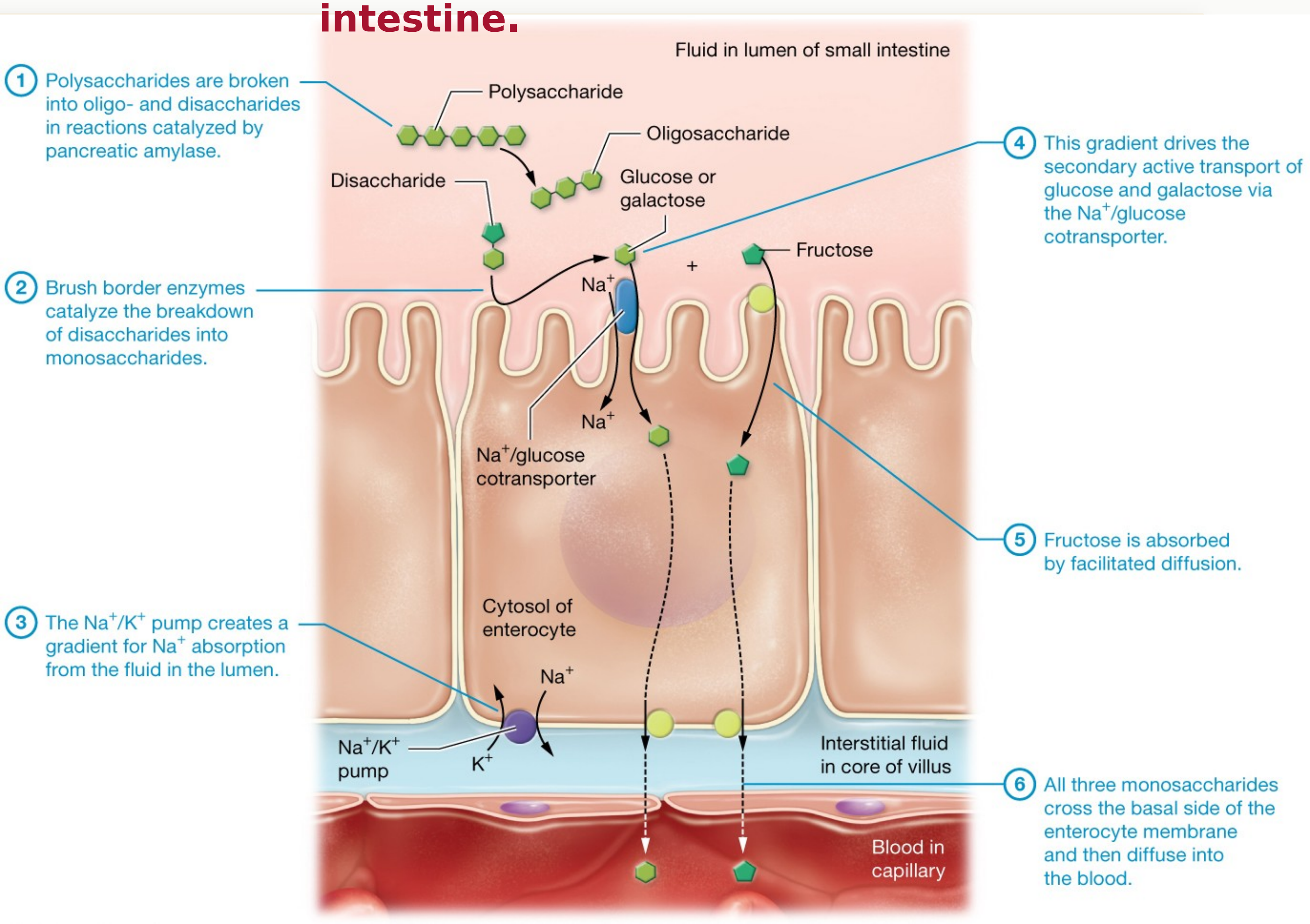
The basolateral Na^+/K^+ ATPase pumps Na^+ OUT of the enterocyte → Na^+ concentration inside the cell stays low.

That low intracellular Na^+ creates a downhill gradient for Na^+ to flow in from the lumen.

1. **On the apical membrane, co-transporters exploit that Na^+ gradient to pull in:**
   - Glucose and Galactose (via SGLT1)

That is secondary active transport: no ATP at the apical membrane itself, but the whole thing runs because the pump keeps the gradient alive.

- Fructose is absorbed through **facilitated diffusion**, a passive process requiring a **concentration gradient**.



The Na⁺ Gradient Is the Engine

CONCEPT

One pump powers the whole absorption system.

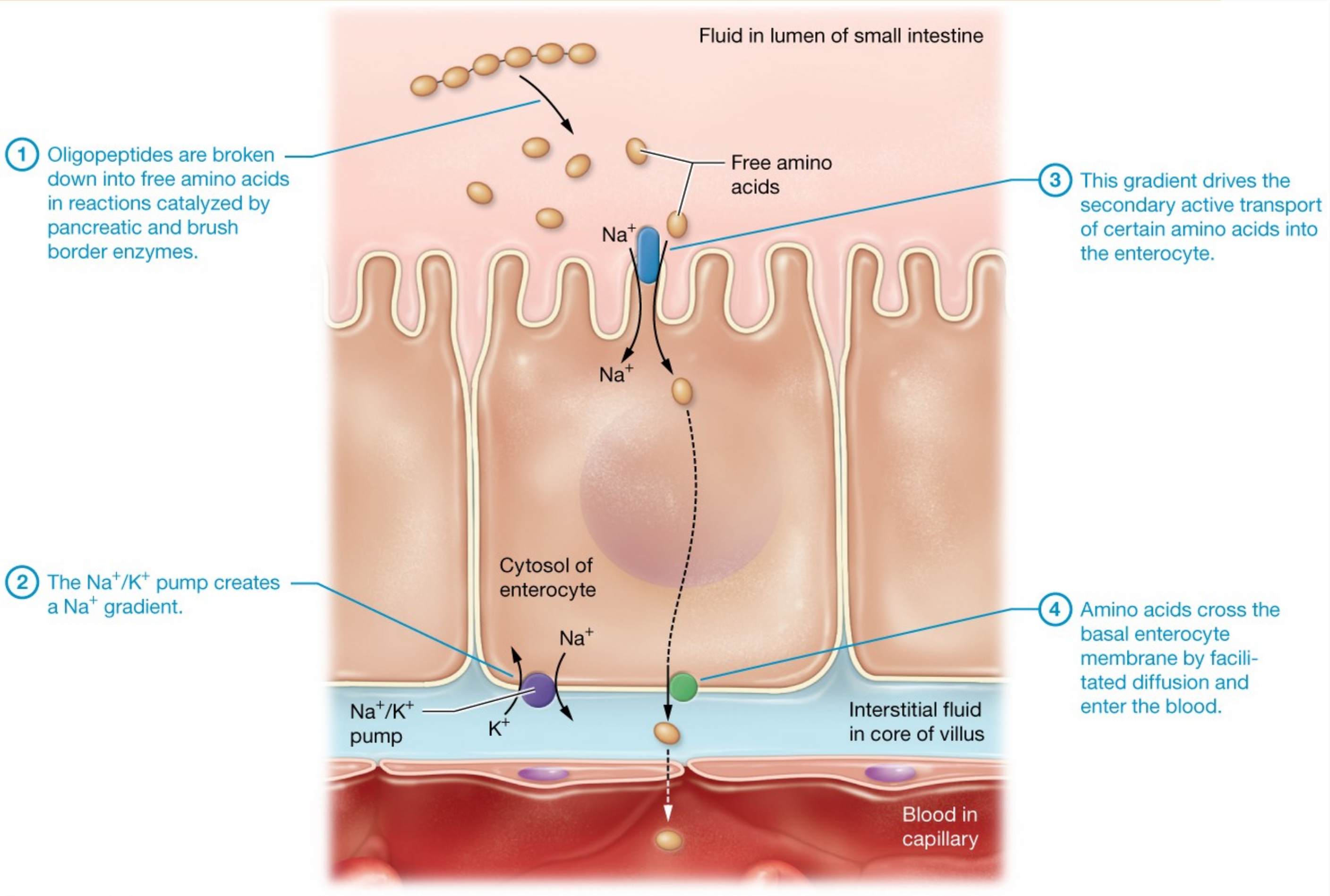
The basolateral Na^+/K^+ ATPase pumps Na^+ OUT of the enterocyte → Na^+ concentration inside the cell stays low.

That low intracellular Na^+ creates a downhill gradient for Na^+ to flow in from the lumen.

1. **On the apical membrane, co-transporters exploit that Na^+ gradient to pull in:**
    - **Amino acids (via Na^+-dependent amino acid transporters)**

That is secondary active transport: no ATP at the apical membrane itself, but the whole thing runs because the pump keeps the gradient alive.

Example: amino acid digestion and absorption in the small intestine.



Activity: Transport Bingo

ACTIVITY

Goal

For each nutrient, identify the transport mechanism at the APICAL membrane and at the BASOLATERAL membrane of the enterocyte.

How it works (electronic)

Groups of three. One laptop or tablet per group.
Open **Transport Bingo: Enterocyte Absorption** from '***Assignments and Activities for Weeks 0 & 1***' (the link is at the end of '***Class Materials: Sessions 1-4***'; download and open in your browser).
Enter a team name. For each nutrient, pick the apical and basolateral mechanism from the dropdown.
Hit **Check Answers:** correct cells turn green, wrong ones turn pink with a hint. Iterate until you get BINGO.

1. Time: 6 minutes. First two groups to BINGO get bragging rights.

Transport Bingo: The Nutrient List

ACTIVITY

Seven nutrients to place

- Glucose
- Amino acids
- Fructose
- Fatty acids + monoglycerides
- Water
- Ions (Na^+, K^+, Ca^{2+}, Fe^{2+})
- Vitamin B12

In the interactive, for each row:

1. Pick the mechanism at the **apical** membrane (lumen → cell).

2. Pick the mechanism at the **basolateral** membrane (cell → blood or lymph).

3. Click Check Answers as often as you like. Use the hint text to fix wrong guesses.

Challenge: which of these would stall if we blocked the Na^+/K^+ ATPase?

Transport Bingo — Debrief

CHECK

Nutrient	Apical	Basolateral
Glucose	Na^+ co-transport (SGLT1) — secondary active	Facilitated diffusion (GLUT2)
Amino acids	Na^+ co-transport — secondary active	Facilitated diffusion
Fructose	Facilitated diffusion (GLUT5)	Facilitated diffusion (GLUT2)
Fatty acids + MAG	Simple diffusion (lipophilic)	Exocytosis as chylomicrons → lacteal
Water	Osmosis	Osmosis
Ions	Various channels / transporters	Various pumps + channels
Vitamin B12	Endocytosis (with intrinsic factor)	Exocytosis

Key insight: the Na^+/K^+ ATPase on the basolateral side powers glucose and amino acid uptake on the apical side.

After Absorption: Two Routes Out of the Villi

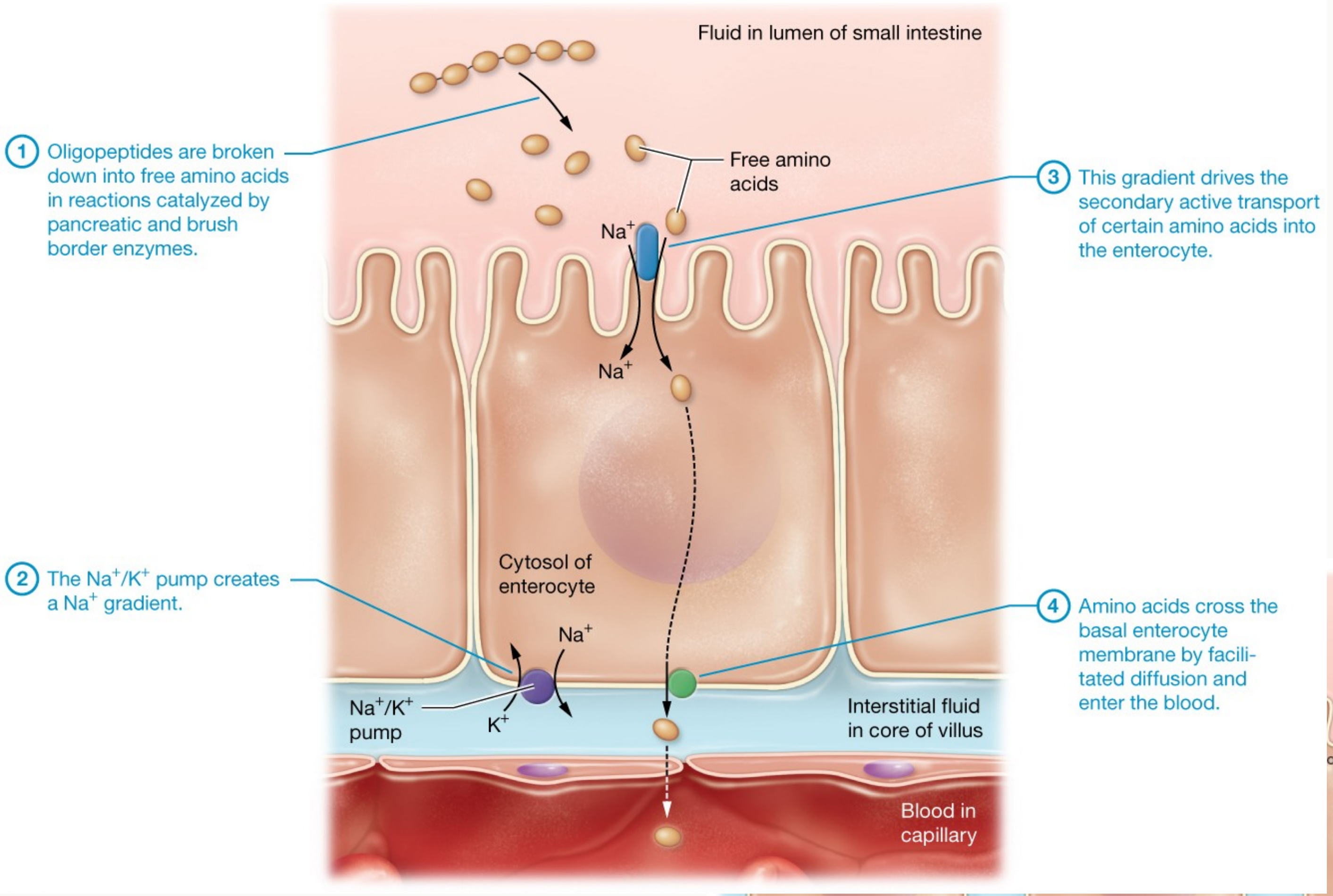
CONCEPT

Route 1 — Capillaries

→ hepatic portal vein → liver → systemic circulation

Who takes this route:
- Monosaccharides (glucose, fructose, galactose)
- Amino acids
- Water-soluble vitamins (B, C)
- Short-chain fatty acids
- Most minerals

The liver gets first look: detoxification + nutrient regulation before the rest of the body sees these molecules.



After Absorption: Two Routes Out of the Villi

CONCEPT

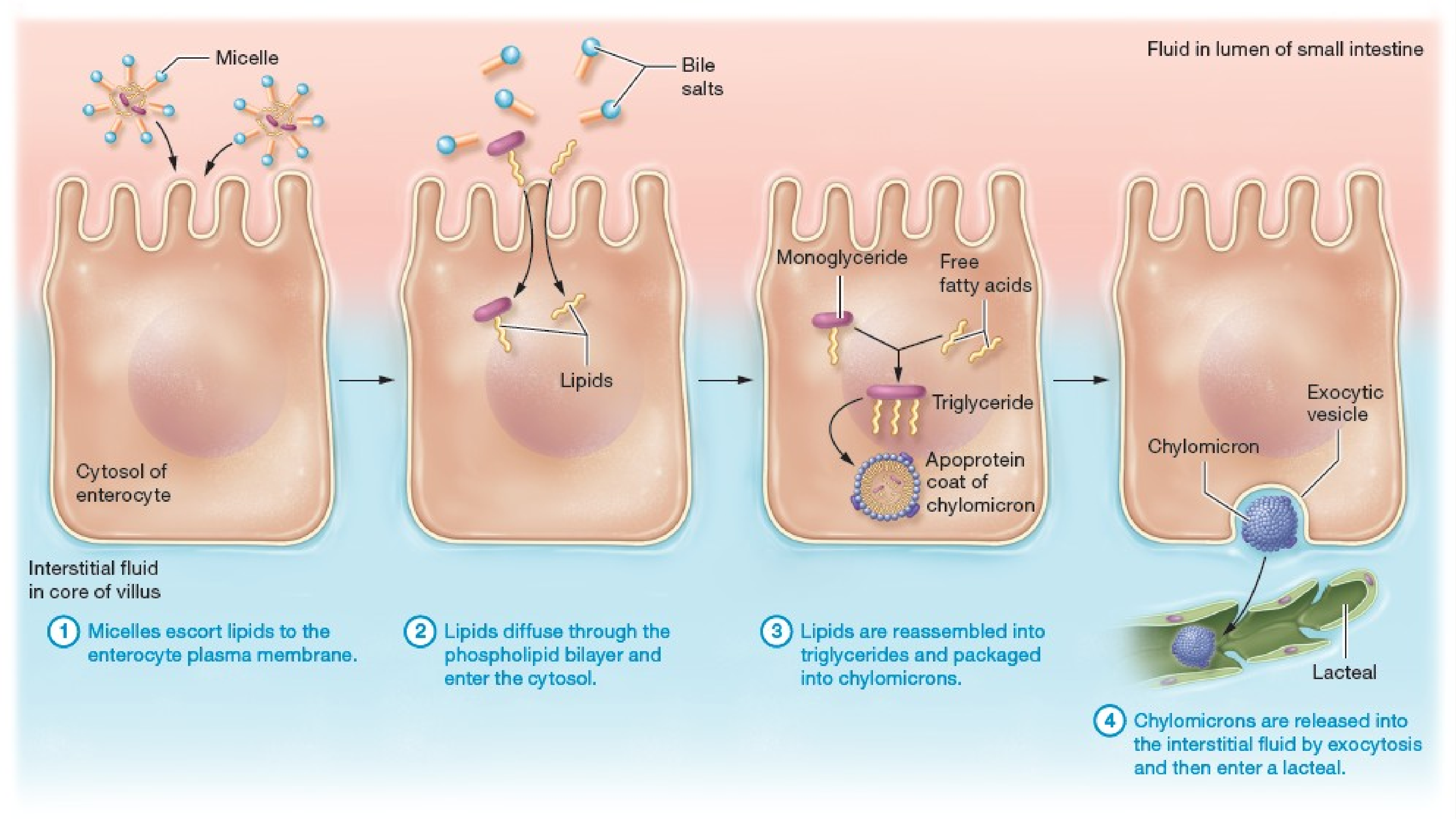
Route 2 — Lacteals (lymph)

→ thoracic duct → left subclavian vein → heart

Who takes this route:
• Long-chain fatty acids (as chylomicrons)
• Fat-soluble vitamins: A, D, E, K

Why lymph, not blood?
Chylomicrons are too large to squeeze into capillary pores. Lacteals are leakier and accept them.

These bypass the liver on the first pass.



Quick Poll: The Olive Oil Question

CHECK

A patient eats a meal high in olive oil. Where do most of the absorbed fat molecules go FIRST?

A) Hepatic portal vein

B) Lacteals and lymphatics

C) Directly to the heart through the aorta

D) They stay inside the enterocyte

A patient eats a meal high in olive oil. Where do most of the absorbed fat molecules go FIRST?

0

Option	Result
Hepatic portal vein	0%
Lacteals and lymphatics	0%
Directly to the heart through the aorta	0%
They stay inside the enterocyte	0%

Answer: Lacteals and Lymphatics

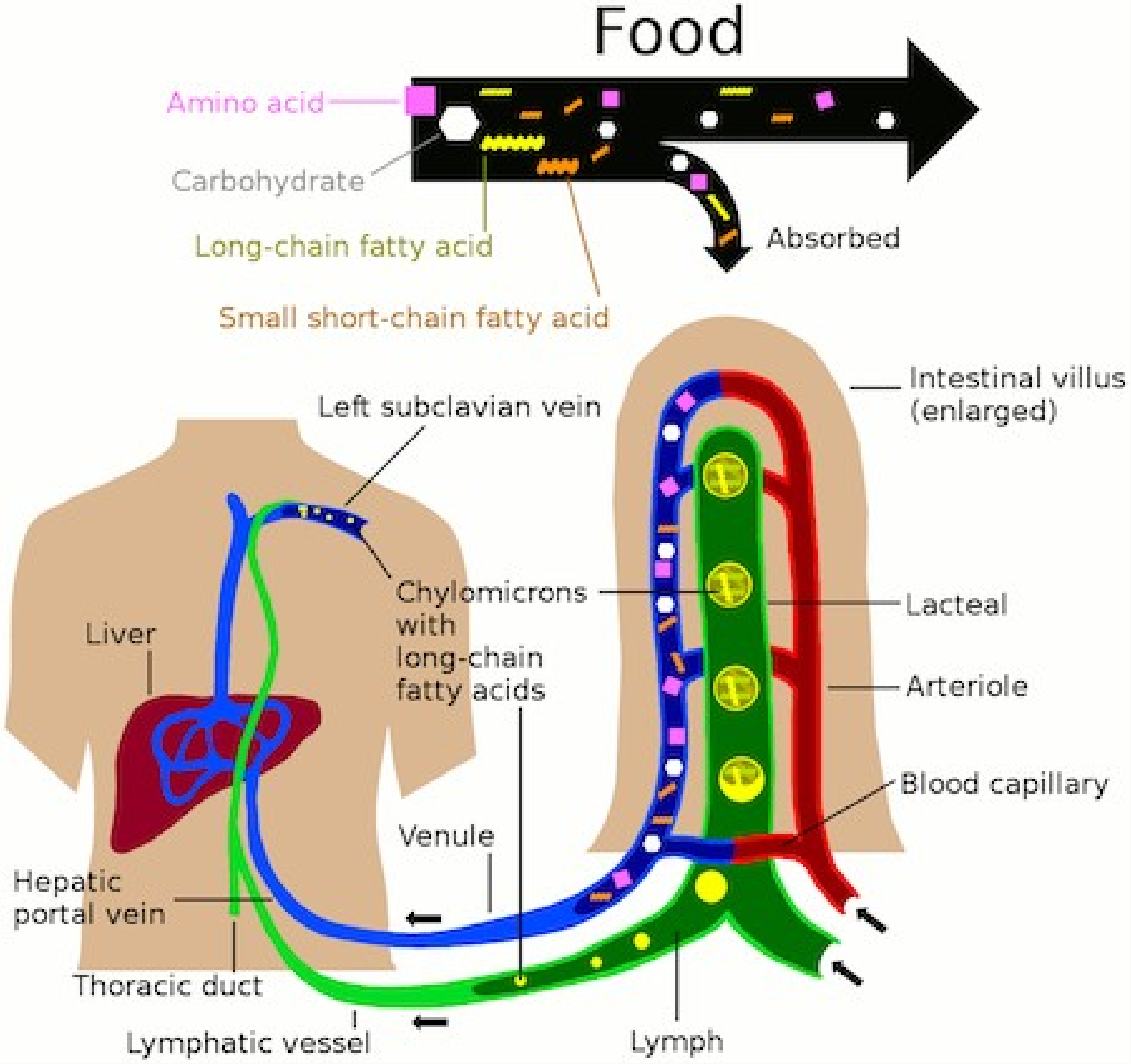
CHECK

✓ B — Long-chain fats ride the lymph first.

Olive oil → triglycerides → emulsified by bile → cleaved by pancreatic lipase → monoglycerides + free fatty acids → absorbed by simple diffusion → reassembled inside the enterocyte → packaged into chylomicrons → exocytosed into the LACTEAL.

From there: lymphatics → thoracic duct → left subclavian vein → right heart → pulmonary circuit → left heart → systemic circulation.

This is why fats bypass the liver on the first pass. The liver will see them later, through systemic circulation.



A&P Preview: Why Lacteals?

A&P REVIEW

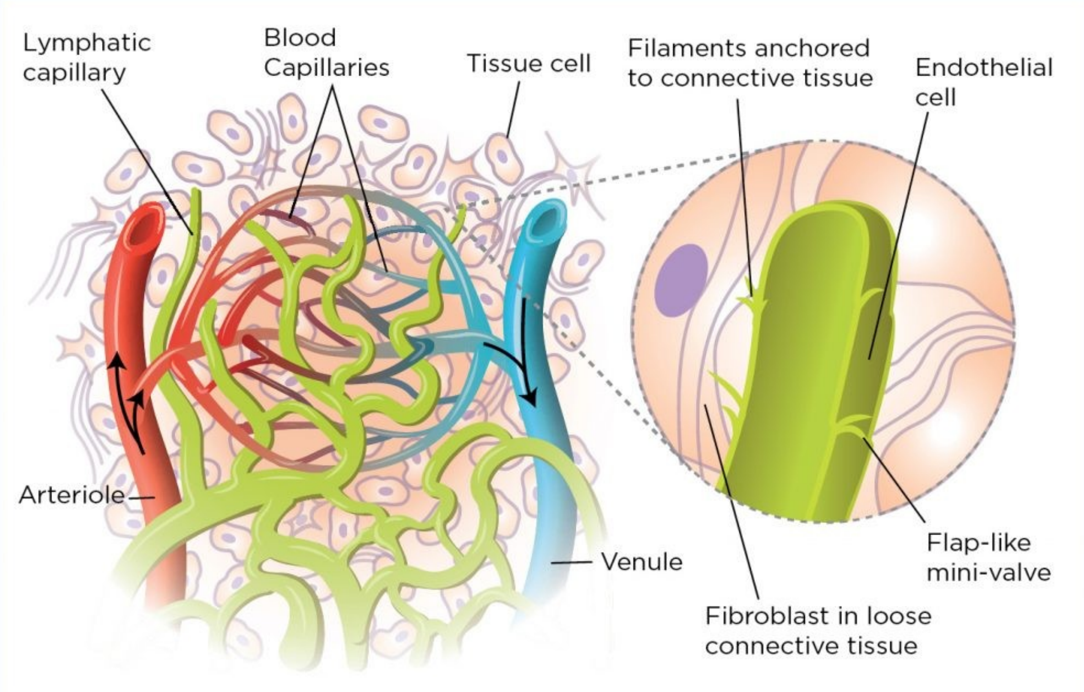
Blood capillaries

- Thin-walled (single endothelial layer)
- Small gaps between cells
- Great for small molecules (glucose, amino acids, ions)
- Too tight for chylomicrons (~75–1200 nm)

Lacteals (lymph capillaries)

- Overlapping endothelial 'flaps'
- Much more permeable
- Accept large particles (chylomicrons, immune cells)
- Drain into the thoracic duct

Return to this in Week 6 — the lymphatic system is not only about immunity. It is a parallel transport highway.



Clinical Tie-In: If You Can't Absorb Fat...

CONCEPT

...you can't absorb fat-soluble vitamins either.

Fat-soluble vitamins: A, D, E, K

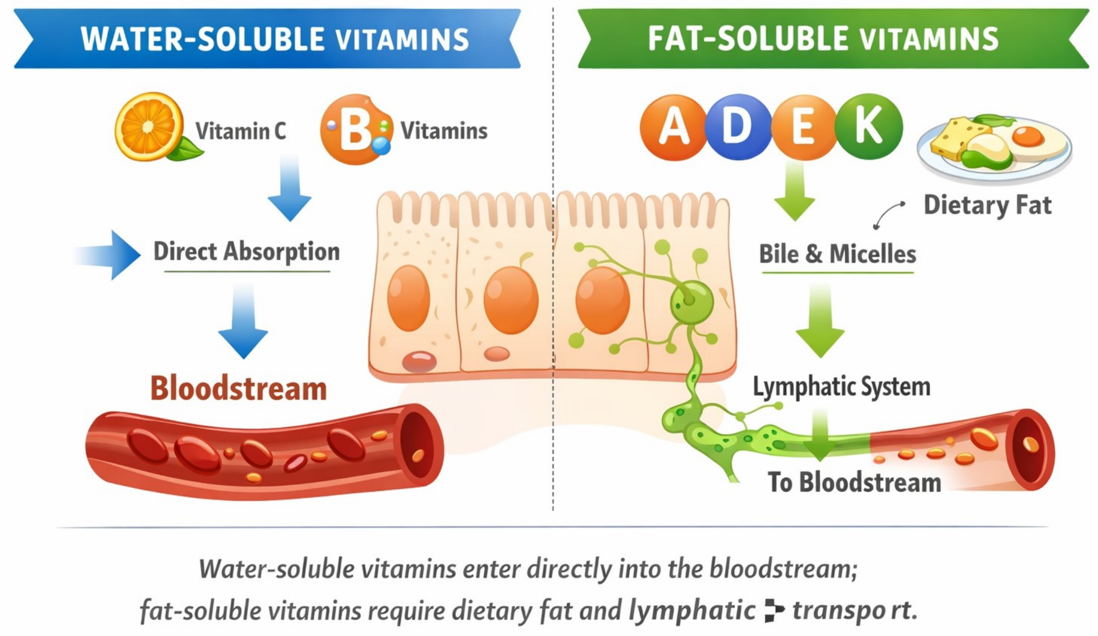
They hitch a ride with dietary fats → need bile, need lipase, need an intact absorption route

Scenarios that break the fat-absorption pipeline:

• Gallbladder removed or bile duct obstructed → poor emulsification
• Pancreatic insufficiency (cystic fibrosis, chronic pancreatitis) → no lipase
• Celiac disease or Crohn's → damaged absorptive surface

Predicted deficiencies:

• Vitamin A → night blindness
• Vitamin D → bone demineralization
• Vitamin E → neurological symptoms
• Vitamin K → bleeding and bruising



BREAK

10 minutes

When we come back: the large intestine and the Coffee capstone.

BLOCK 3

Large Intestine

Water, microbiome, and compaction

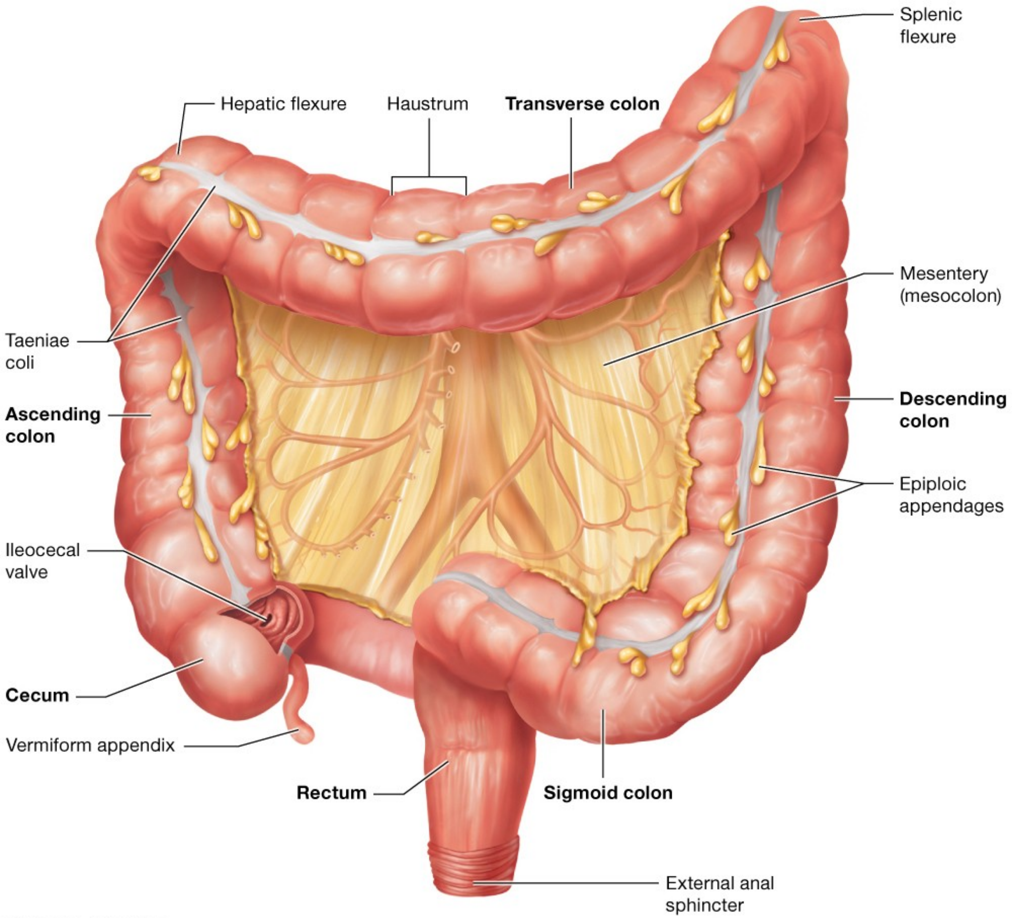
Large Intestine: Anatomical Tour

CONCEPT

Regions (follow the flow)

1. Cecum (+ appendix)
2. Ascending colon
3. Transverse colon
4. Descending colon
5. Sigmoid colon
6. Rectum
7. Anal canal

Total length: ~1.5 m. Much shorter than the small intestine.



(a) Large intestine

Three Features You Won't Find in the Small Intestine

CONCEPT

Teniae coli

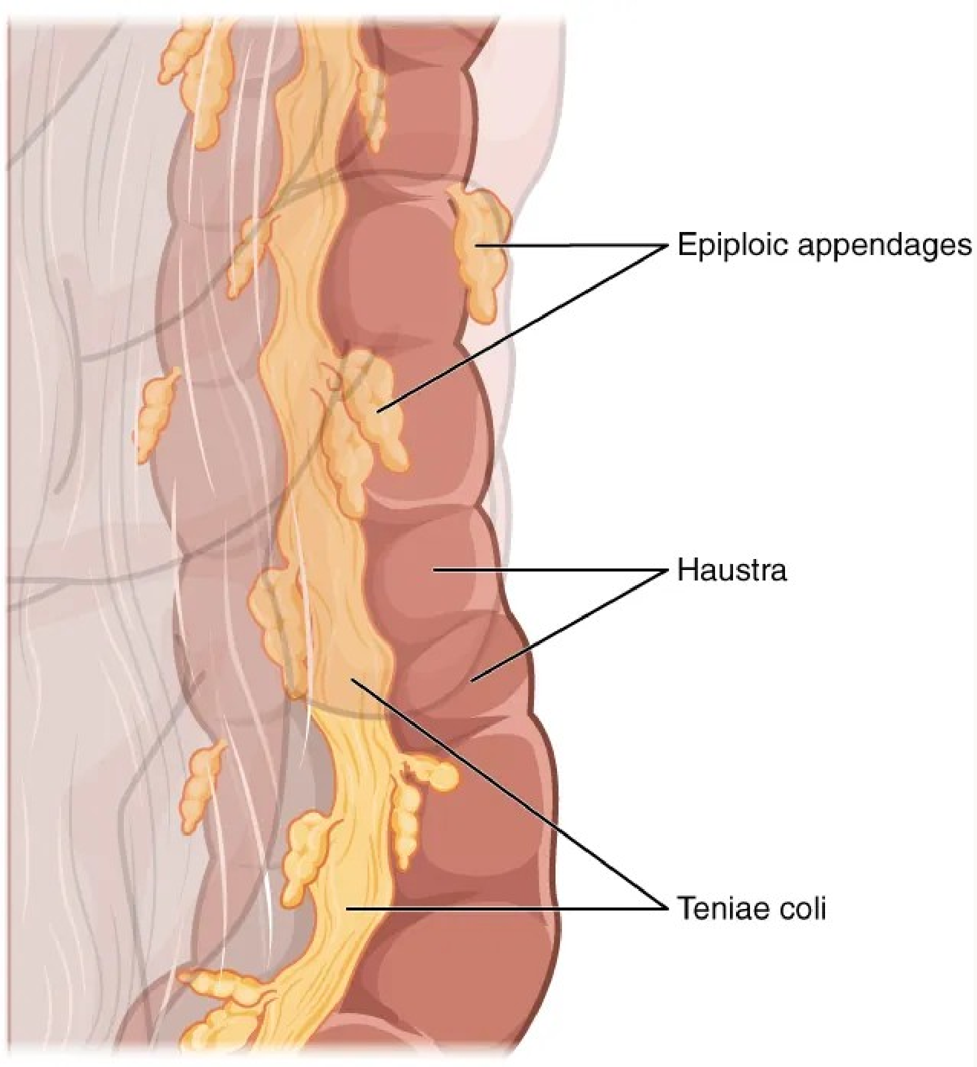
Three longitudinal bands of smooth muscle running the length of the colon. They gather the wall into pouches.

Haustra

The pouches between teniae coli contractions. Responsible for the colon's segmented appearance.

Epiploic appendages

Small fat-filled sacs hanging off the outer wall. Function is still debated.



What the Large Intestine Actually Does

CONCEPT

Absorb water and electrolytes

Reclaims most of the water still in chyme (~1 L/day). Also Na^+ and Cl^-.

Host the microbiome

Trillions of bacteria ferment indigestible carbs → short-chain fatty acids, gas, vitamin K, some B vitamins.

Compact waste

Dehydrated, formed feces ready for elimination.

Defecation reflex

Rectal distension → involuntary relaxation of the internal anal sphincter; voluntary control over the external sphincter.

The Microbiome: Your Second Genome

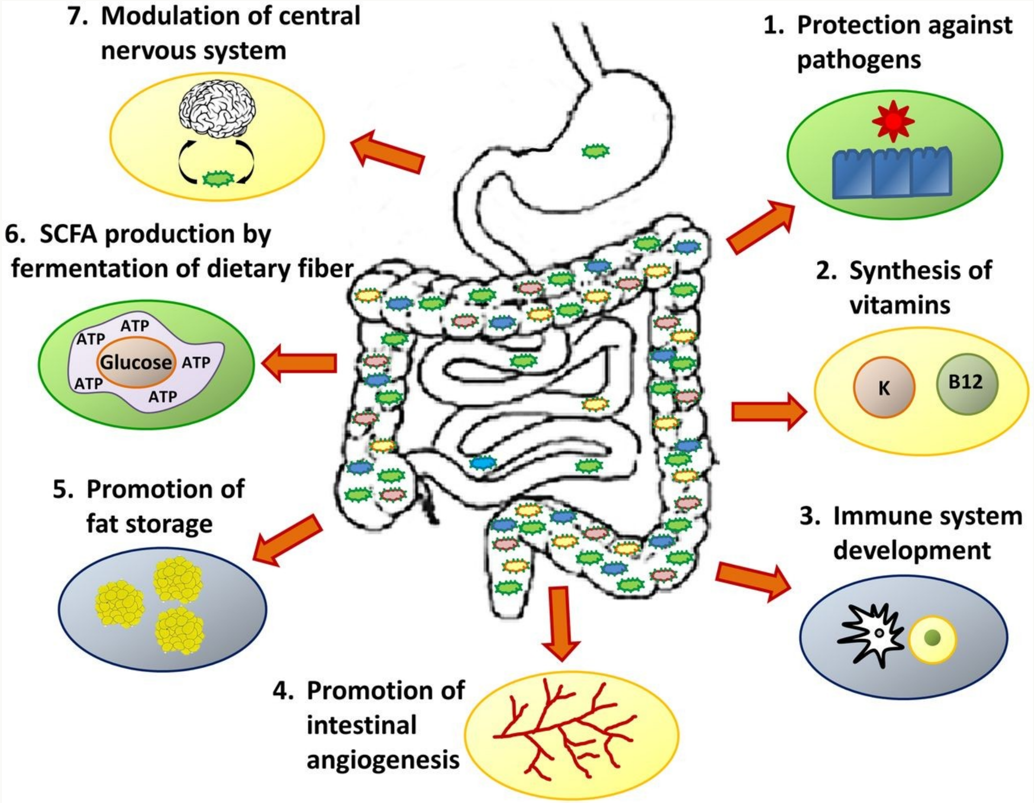
CONCEPT

~10^{13}–10^{14} bacteria in your colon. They outnumber your own cells.

What they do for you:
• Ferment indigestible fibers into short-chain fatty acids (butyrate, propionate, acetate) → fuel for colonocytes
• Synthesize vitamin K and several B vitamins
• Compete with pathogens for space and nutrients (colonization resistance)
• Educate and tune the immune system

What disrupts them:
• Broad-spectrum antibiotics → can trigger C. difficile overgrowth
• Chronic low-fiber diets → reduced diversity
• Inflammatory bowel disease → altered composition

Fast-moving research area: microbiome connections to metabolism, mood, immunity, even cancer.



Think – Pair – Share

ACTIVITY

Prompt

A patient has severe damage to the large intestine from ulcerative colitis. Predict three symptoms and explain WHY each one occurs — using the functions we just listed.

Structure

Think — 60 seconds alone
Pair — 90 seconds with a neighbor
Share — cold call two pairs

Debrief: Ulcerative Colitis Symptoms Explained

CHECK

Watery diarrhea

Water absorption fails → stool stays liquid

Electrolyte imbalances

Na^+, K^+, Cl^- lost with the fluid

Bleeding / anemia

Ulcerated mucosa bleeds into the lumen

Easy bruising

Vitamin K deficiency → impaired clotting

Altered microbiome

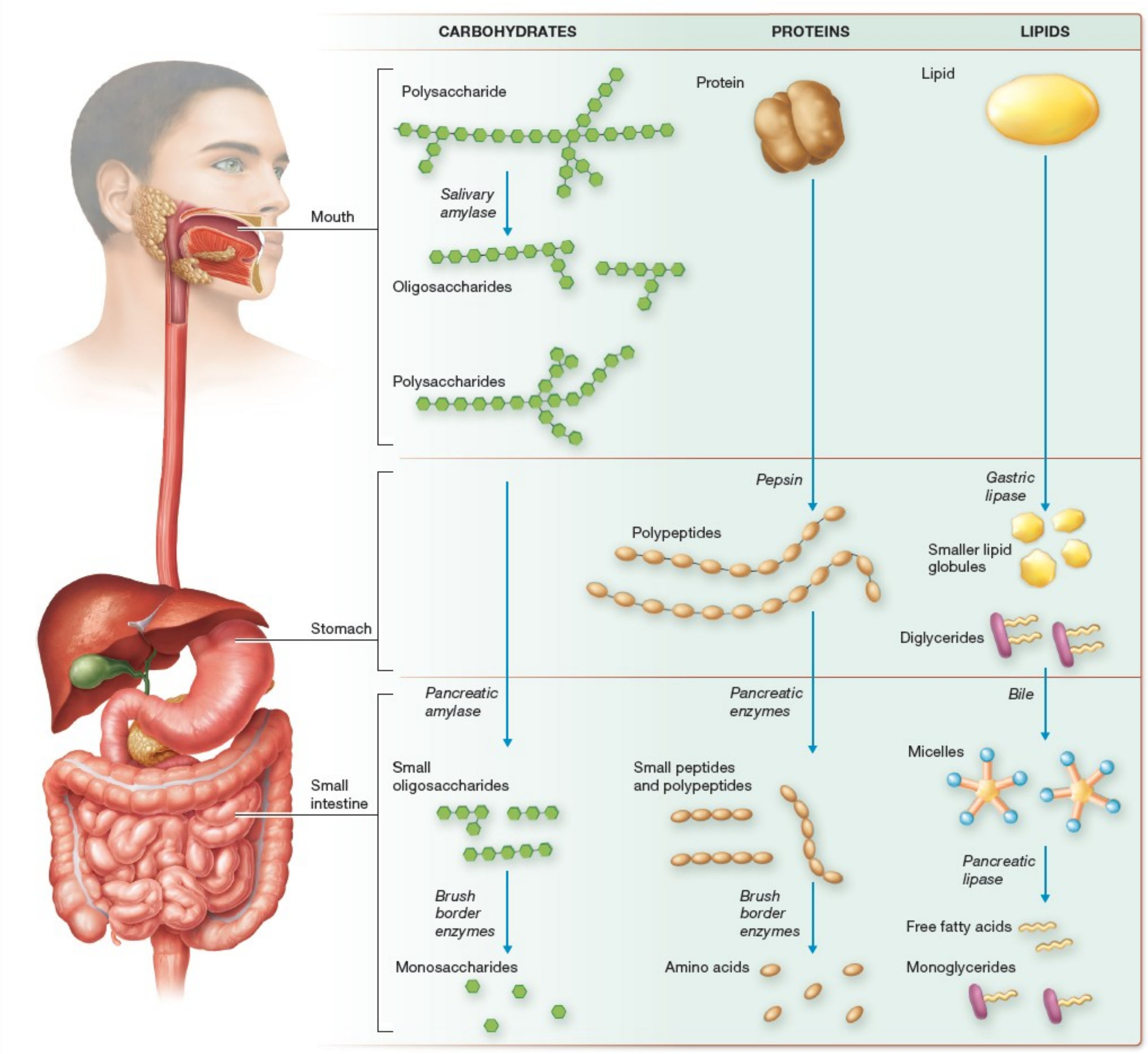
Inflammation disrupts bacterial balance

Weight loss / fatigue

Chronic inflammation + malabsorption

The Complete Digestive Picture

CONCEPT



BLOCK 4

Integration Capstone

Coffee with Cream and Sugar: Take-Home Application Activity

Application Activity: Coffee with Cream and Sugar

ACTIVITY

The prompt

You drink a cup of coffee with cream and sugar.

Trace what happens to the **water, the sugar (sucrose), the milk fat, and the milk protein** from your lips to the bloodstream (or lymph).

At every major stop, describe:

what is happening mechanically and chemically

which organ or cell is responsible

- which transport mechanism moves each nutrient out of the GI tract

Logistics

Where: Camino → Weeks 0 & 1 module → Application Activity: Ingestion to Absorption

Format: one short write-up per student (solo). A diagram or table is encouraged.

Points: 10 pts

Due: Saturday, April 18 at 11:59 PM

Tip: *pull directly from Transport Bingo: the seven nutrients we just placed are the answers to most of this.*

The Journey: Six Stops

ACTIVITY

1	2	3	4	5	6
Mouth	Esophagus	Stomach	Small intestine	Absorption site	Destination

At Each Stop, Answer Four Questions

ACTIVITY

1 **Mechanical digestion?**

What physical process is happening (chewing, churning, segmentation)?

2 **Chemical digestion?**

What enzyme is cleaving what bond? Name the enzyme.

3 **Absorption mechanism?**

At which membrane, by what transport mechanism?

4 **Destination?**

Portal vein (liver first) or lacteal (lymph first)?

Debrief: Common Insights

CONCEPT

Sugar is two sugars

Sucrose = glucose + fructose. Sucrase at the brush border cleaves them. GLUT5 for fructose, SGLT1 for glucose.

Cream is a mini meal

Fat + protein + lactose. Different enzymes, different routes. One beverage demonstrates the whole unit.

Caffeine takes a shortcut

Small, lipophilic. Simple diffusion in the stomach and upper small intestine. Straight to the blood.

Water follows solutes

Once Na^+ and glucose are absorbed, water follows by osmosis. This is why oral rehydration salts contain sugar.

Why This Actually Saves Lives

CONCEPT

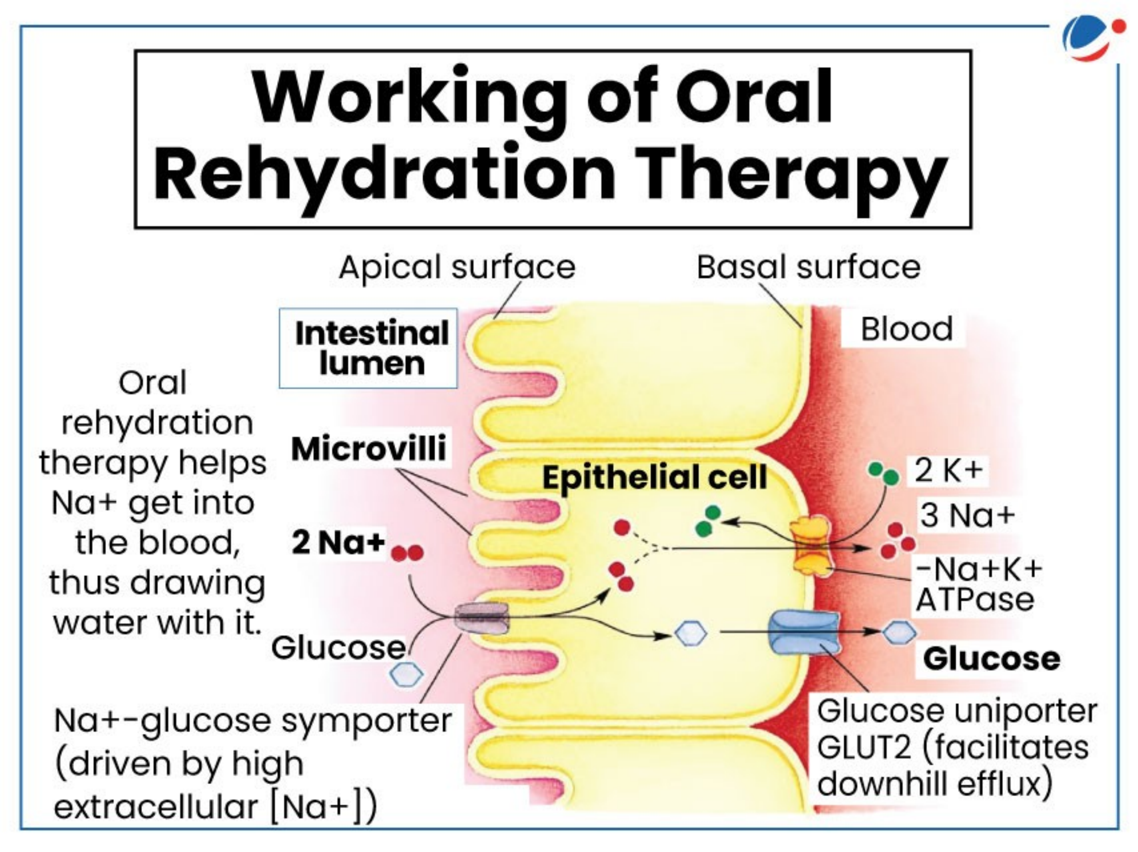
Oral Rehydration Solution (ORS)

In cholera, enterocytes lose huge amounts of water into the lumen. Patients can die from dehydration in hours.

The treatment is a simple salt + sugar + water drink. Why does adding sugar matter?

Because SGLT1 co-transports Na^+ WITH glucose. If you give Na^+ alone, the transporter does not move. Add glucose — Na^+ moves in, water follows by osmosis, the patient rehydrates.

One co-transporter. Millions of lives saved. This is physiology at its most practical.



Connecting to Your Assignments

CONCEPT

Application Activity — Ingestion to Absorption

You just did a version of this in class. The assignment asks you to pick a different food and trace it through the system. Use your class notes and the textbook.
Due Saturday, April 18.

RCS Quiz 1 — Running Case Study

A patient with a digestive problem. You will need to apply anatomy, histology, enzymes, transport, and regulation. The REASONING matters more than memorizing every enzyme name.
Due Saturday, April 18.

How to Read an RCS Case

CONCEPT

1	**Read the case twice**	First for story. Second to underline anatomical clues and symptoms.
2	**Name the location**	Where in the GI tract is the problem? Be specific.
3	**State normal function**	What is that structure supposed to do for digestion and absorption?
4	**Explain the break**	How does the disease disrupt that normal function? Trace the consequence.
5	**Predict symptoms**	What symptoms would you expect from that disruption? Match them to the case.

Looking Ahead to Week 2

CONCEPT

Metabolism and Nutrition

You just learned how nutrients enter the body. Next week: what happens to them once they are inside.

Coming topics:
- Glycolysis, Krebs cycle, oxidative phosphorylation
- Fat and protein metabolism
- Absorptive vs. post-absorptive states
- Metabolic rate and nutrition basics

Heads up:
- ALW for Week 2 (Metabolism) due Sunday, April 19
- RCS Quiz 1 and Application Activity due Saturday, April 18

Molecules → Cells → Tissues → Organs → You

Today you used every level, at every scale.

Great work these two weeks. The digestive system is your foundation — everything else builds on it.

Questions? Office Hours This Week

CONCEPT

Where to get help

- Footsie and AIDA: You get help 24/7 AND Extra Credit!
- Office hours: after class
- Q&A board on Canvas — post anytime
- Tutoring Center drop-in hours
- Study groups — form your own or ask me to help pair you up

Before the next class, make sure you:

- Finish RCS Quiz 1 + Application Activity (Sat)
- Start Week 2 ALW (Sun)
- Read the metabolism chapter preview